# Targeted Thesis Revision Guide

## Duplicate Detection, Identity Matching and Scholarship Status

**Purpose:** Identify only the paragraphs that need correction for the six earlier questions: compared fields, checking order, Levenshtein usage, thresholds, scholarship status, and the effect of a possible duplicate. This is not a rewritten thesis.

**Documents reviewed:** the original 86-page thesis PDF and Summary_Comments_Scholarship_System_Final_Defense.docx. Page references below refer to the original thesis, whose printed numbers match its PDF page positions, not this guide.

**Adviser alignment:** Comment 4 is the primary basis (fields, threshold, handling and tests). Comments 5, 9, 2 and 12 apply to testing evidence, external-record coverage, objective/finding consistency and repeated terminology. Unrelated recommendation-model, survey, privacy and formatting revisions are outside this focused guide.

| Edit | Original Thesis Location | Purpose |
|---|---|---|
| R1 | Chapter II - Technical Background / Duplication Prevention, pp. 21-22 | Correct the short implementation explanation. |
| R2 | Chapter III - System Modules / 6. Duplication Prevention Algorithm, pp. 41-42 | State fields, formulas and exact thresholds. |
| R3 | Chapter III - 7. Approval Module and 8. Monitoring Module, p. 42 | Separate approval types; explain active vs. closed. |
| R4 | Chapter III - 5. Validation Module, pp. 40-41; Requirements Analysis, p. 50; Chapter IV, pp. 56-57 | Correct repeated automatic-blocking claims. |
| R5 | Chapter III - Testing, pp. 33-34 | Identify evidence needed; do not invent passes. |
| R6 | Chapter I - Introduction, p. 6; Objectives, p. 8; Conceptual Framework, p. 11; Limitations, p. 14 | Qualify guarantees and explain data coverage. |
| R7 | Chapter V - Finding 2, pp. 79-80; Conclusions, p. 81; Recommendations, pp. 82-83 | Align final claims with the verified behavior. |

**Evidence boundary:** Replacement text is based on inspected local source and migration definitions, not a fresh production test. Thresholds are configured rules, not measured accuracy or proof that every duplicate is detected. Original files remain unchanged.

## R1. Correct the Technical Background Explanation

**Exact location:** Chapter II - Review of Related Literature and Studies > Technical Background > Duplication Prevention, pp. 21-22

**Find the paragraph beginning:** "The duplication prevention process may"

**Stop at:** "manual review."

**Edit boundary:** Replace the last three implementation paragraphs, starting with the exact-matching paragraph on p. 21 and ending with the threshold paragraph on p. 22. Keep the introductory paragraph and the Record Linkage definition/citation. Do not replace Technology Acceptance Model, which follows on p. 22.

**Problems:** This passage lists birthdate and school, describes fuzzy matching only as a fallback, leaves the threshold unspecified, and implies every possible duplicate prevents approval. Adviser comment 4 directly requests these corrections.

**Suggested replacement / insertion:**

The system uses separate identity-matching and scholarship-eligibility checks. During registration, roster records are located using the submitted Student ID and the associated name is checked for consistency. During roster import, exact student identity, existing authoritative roster assignments and scholarship commitments are validated. The weighted duplicate matcher is a separate record-comparison function; it evaluates the records supplied to it rather than independently querying only active scholarships.

The weighted matcher compares Student ID (32%), full name (30%), email (10%), contact number (8%), course (8%), year level (4%), city or municipality (4%), and province (4%). Birthdate and school are not comparison fields in this implementation. Address matching within this score is limited to city and province. Levenshtein similarity is calculated for available comparison values even when the Student IDs match. Unavailable comparisons are excluded and the remaining weights are normalized.

The general fuzzy-match rule requires at least 84% weighted similarity and 82% name similarity. An exact Student ID is independently sufficient to identify a duplicate; an exact email or contact number also identifies a duplicate when name similarity is at least 72%. The administrative grouping audit additionally uses a default 84% score threshold. A suspected match supports record review, whereas backend roster-conflict and commitment checks restrict scholarship operations. Account activation is separate from scholarship assignment and is not universally blocked by a suspected duplicate.

**Code reference:** backend/grantor_algorithms.py:68-120, 206-215; backend/signup_service.py:519-586; backend/roster_workflow_service.py:230-292

**Consistency instruction:** Use R2 for the detailed method. Do not insert the old field list or the unspecified 'defined threshold' again in Chapter III.

## R2. Replace the Algorithm's Fields, Formula and Rules

**Exact location:** Chapter III - System Modules > 6. Duplication Prevention Algorithm, pp. 41-42

**Find the paragraph beginning:** "The duplication prevention process begins with exact matching."

**Stop at:** "only on a newly created rule-based process."

**Edit boundary:** Replace the whole body of Section 6, including its existing weighted-score formula and the threshold paragraph on p. 42. Keep the Section 6 title and stop before 7. Approval Module. This resolves questions 1-4 and adviser comment 4.

**Suggested replacement / insertion:**

The duplicate-comparison function implements weighted record linkage with normalized Levenshtein similarity. It normalizes case, punctuation and whitespace before comparison. Names are compared in their original and token-sorted order, and the higher similarity is used. Exact identifier rules and fuzzy similarity rules are evaluated together; fuzzy comparison is not restricted to missing or inconsistent Student IDs.

| Comparison Field | Weight |
|---|---|
| Student ID | 0.32 |
| Full name | 0.30 |
| Email | 0.10 |
| Contact number | 0.08 |
| Course | 0.08 |
| Year level | 0.04 |
| City/municipality | 0.04 |
| Province | 0.04 |

**Use these formulas:** S(a,b) = 1 - d(a,b) / max(len(a), len(b)); W = sum(w_i x S_i) / sum(w_i), using only included comparisons. Here d is Levenshtein edit distance, S_i is field similarity and w_i is its configured weight. If no comparisons are available, W is zero.

The function identifies a duplicate when: (1) Student IDs match exactly; or (2) email or contact number matches exactly and name similarity is at least 0.72; or (3) name similarity is at least 0.82 and W is at least 0.84. The administrative grouping audit also groups records when W reaches its default 0.84 threshold, without necessarily requiring the 0.82 name condition.

Registration uses a separate roster-name check after an exact Student ID lookup: direct or token-sorted name similarity must reach 0.72, or matching last names must have at least 0.50 token overlap. Student ID is mandatory during registration. These configured thresholds are not measured accuracy rates.

**Code reference:** backend/grantor_algorithms.py:10-65, 68-120, 206-215; backend/signup_service.py:519-586

## R3. Explain Active, Pending and Closed Scholarships

**Exact location:** Chapter III - System Modules > 7. Approval Module, p. 42

**Find the paragraph beginning:** "The approval module provides scholarship grantors"

**Edit boundary:** Replace only the body paragraph of 7. Approval Module; retain the heading. The old text treats approval as one action and says other applicant lists may be removed or disabled without explaining when the commitment occurs.

**Suggested replacement / insertion:**

Account approval, document review and scholarship assignment are separate operations. Account approval may activate a student while a conflicting roster assignment is recorded for authorized resolution. Scholarship operations remain subject to backend commitment and roster-conflict checks. Students may hold uncommitted applications with different grantors, but another open application under the same grantor is restricted; repeating the same application returns the existing record. A scholarship commitment, including the commitment established by requesting materials, prevents a second commitment and closes competing applications through the transaction.

**Exact location:** Chapter III - System Modules > 8. Monitoring Module, p. 42

**Find the paragraph beginning:** "This module enabled continuous tracking"

**Edit boundary:** Keep the existing monitoring description. Insert the following status clarification immediately after it and before 9. Notification Module.

| State | Clarification to Insert |
|---|---|
| Active / committed | An existing commitment or authoritative assignment restricts another scholarship. |
| Pending / uncommitted | Different-grantor applications may coexist; same-grantor open-application restrictions still apply. |
| Finished / completed requirements | Completion may mean the semester's requirements are finished; it does not automatically release the scholarship commitment. |
| Closed records | The shared open-application check excludes rejected, denied, declined, cancelled/canceled, withdrawn, archived, resolved and expired records. Other restrictions and stored commitments must also be checked. |
| Rejection / archive | Rejection can impose a 24-hour same-grantor cooldown. An individual archive can require an invitation before reapplication to that grantor. |
| Inactive / terminated | These labels are not uniformly implemented as terminal scholarship states. Do not state that either automatically permits another application. |

**Code reference:** supabase/migrations/20260906124238_multiple_applications_one_choice.sql:4, 130-205; 20260918130000_automatic_roster_and_login_security.sql:299; 20261010154625_allow_account_approval_with_roster_resolution.sql:90

## R4. Correct the Repeated Blocking Claims

**Problem:** A fuzzy match, a roster conflict and an existing commitment are not interchangeable. Adviser comments 4 and 12 require the repeated module descriptions to agree. Make the three limited replacements below.

**Exact location:** Chapter III - System Modules > 5. Validation Module, pp. 40-41

**Find the paragraph beginning:** "The validation module serves as a core component"

**Edit boundary:** Replace both paragraphs under Section 5, stopping before 6. Duplication Prevention Algorithm. Leave the recommendation section above intact.

**Suggested replacement / insertion:**

The validation module combines identity checks, duplicate-record comparison and scholarship-commitment validation. Weighted Levenshtein matching supports administrative duplicate review and grantor record editing. The current roster import workflow checks exact student identity, normalized-name consistency, active roster assignments and existing commitments; identity or assignment conflicts block the import and are recorded for resolution. Backend checks also restrict scholarship operations when unresolved roster conflicts or existing commitments are present. A fuzzy audit result does not universally prevent account activation. Coverage is limited to internal records and external records actually encoded or imported into the system.

**Exact location:** Chapter III - Requirements Analysis > 5. Validation and Approval Mechanism, p. 50

**Find the paragraph beginning:** "The system implements a pre-approval validation mechanism"

**Edit boundary:** Replace this numbered item's body only.

**Suggested replacement / insertion:**

The system uses identity matching, duplicate-record review and transactional scholarship-commitment checks to support the one-active-scholarship policy. Suspected identity matches and authoritative roster conflicts receive different treatment. Scholarship actions are restricted by unresolved roster conflicts and existing commitments; account activation remains a separate process. Validation is limited to the records available in the system.

**Exact location:** Chapter IV - II. Designing and Developing a Centralized Scholarship Management and Tracking System > The Validation Module, pp. 56-57

**Find the paragraph beginning:** "The Validation Module performs pre-approval checking"

**Edit boundary:** Replace the one paragraph spanning the page break; stop before The Approval Module on p. 57.

**Suggested replacement / insertion:**

The Validation Module provides record-comparison and scholarship-conflict checks. The administrative fuzzy audit groups potential duplicate identities and suppresses redundant Student List entries without deleting records or itself blocking account approval. The current roster workflow records identity and assignment conflicts and blocks conflicting imports. Backend commitment and roster checks restrict affected scholarship operations until the relevant conflict is resolved.

**Code reference:** src/pages/AdminDashboard.jsx:1958-1989; src/pages/ProviderDashboard.jsx:3150-3178, 3255-3320; backend/roster_workflow_service.py:230-340; backend/scholarship_choice_service.py:69-111

## R5. Add Evidence Without Inventing Test Results

**Exact location:** Chapter III - Research Methods > Testing, pp. 33-34

**Find the paragraph beginning:** "Functional testing verified whether each system module"

**Edit boundary:** In this paragraph, replace only the sentence beginning 'The Decision Tree recommendation process and the Record Linkage...' as it concerns duplicate validation. Retain recommendation-test claims only if separate evidence exists. Insert the duplicate-test table before the next evaluation paragraph beginning 'The developed system was also evaluated'. Adviser comments 4 and 5 apply.

**Suggested replacement / insertion:**

Duplicate-validation testing must distinguish the weighted comparison function, registration identity checks, roster import restrictions and transactional scholarship-commitment checks. For each executed case, report the input, expected result, actual result, status and evidence reference. Configured similarity thresholds and source-code inspection alone do not establish accuracy or successful deployment.

**Evidence to collect:** The following are required scenarios, not executed results. Use 'Not yet recorded' for actual results and status until a traceable run is attached.

| Suggested ID | Scenario | Expected Result |
|---|---|---|
| DD-01 | Exact Student ID; other fields differ | Weighted matcher identifies a duplicate via exact ID. |
| DD-02 | Different IDs; spelling variations | Fuzzy result follows both the 82% name and 84% weighted thresholds. |
| DD-03 | Missing ID | Registration rejects missing ID; generic comparison can still evaluate available fields. |
| DD-04 | Threshold boundaries and missing fields | Check just below/at 72%, 82%, 84%; renormalize available comparisons. |
| DD-05 | Existing active commitment | New scholarship action is blocked; account approval is evaluated separately. |
| DD-06 | Pending applications | Different grantors may coexist; same-grantor duplicate is restricted/idempotent. |
| DD-07 | Closed, inactive and terminated labels | Verify each label separately; do not assume inactive/terminated release a commitment. |
| DD-08 | Finished semester requirements | Completion alone does not permit a second commitment. |
| DD-09 | Possible fuzzy duplicate vs. roster conflict | Audit groups records; roster conflict blocks import and records a conflict. |
| DD-10 | Unencoded external scholarship | Document the coverage limitation, not a claim of detecting unseen records. |

**Evidence note:** A roster identity-mismatch test exists in backend/tests/test_roster_workflow_service.py. This matrix was not executed here. Do not infer Pass results from the earlier summary PDF.

## R6. Qualify Only the Relevant Chapter I Claims

These are short consistency edits for adviser comments 2, 4 and 9, not a rewrite of the background, objectives or limitations.

**Exact location:** Chapter I - Introduction, p. 6

**Find the paragraph beginning:** "One major concern in scholarship management"

**Edit boundary:** Keep the literature discussion. Replace only the final sentence starting 'This ensures that duplication is prevented...'.

**Suggested replacement / insertion:**

These checks support the prevention of conflicting scholarship commitments among records available in the system; they do not guarantee detection of every duplicate identity or scholarship held outside its recorded data.

**Exact location:** Chapter I - Objectives of the Study > Specific Objectives > Objective 2, p. 8

**Find the paragraph beginning:** "To develop a system that validates student records"

**Edit boundary:** Replace Objective 2 only; keep its number and the other objectives.

**Suggested replacement / insertion:**

To develop a system that validates student identities, flags possible duplicate records for authorized review, and restricts conflicting active scholarship commitments using internal and externally sourced records encoded or imported into the system.

**Exact location:** Chapter I - Conceptual Framework > process-stage discussion, p. 11

**Find the paragraph beginning:** "After application submission, the system performs"

**Edit boundary:** Replace the sentences from 'After application submission...' through 'another active scholarship.' Retain the final monitoring/reporting sentence.

**Suggested replacement / insertion:**

The system checks available identities, authoritative roster assignments and existing scholarship commitments at the relevant workflow stages. Uncommitted applications may exist under different grantors, but committing to a scholarship restricts another commitment and closes competing applications. Account approval is separate and may activate a student while a roster conflict awaits staff resolution. Authorized users review applications and documentary requirements through their respective workflows.

**Exact location:** Chapter I - Scope and Limitations of the Study > Limitations of the Study, p. 14

**Find the paragraph beginning:** "The system does not provide full real-time system-to-system integration"

**Edit boundary:** Keep the existing external-integration paragraph; it already discloses manual encoding. Append the two sentences below before the accuracy paragraph.

**Suggested replacement / insertion:**

Duplicate and commitment checking covers only records entered or imported into the system. A scholarship absent from these records cannot be detected by the matching process, and similarity thresholds may produce false matches or missed matches.

**Code reference:** backend/grantor_algorithms.py:126-146; backend/signup_service.py:546-586; backend/scholarship_choice_service.py:69-111

## R7. Align Findings, Conclusions and Recommendations

**Exact location:** Chapter V - Summary of Findings > Finding 2, pp. 79-80

**Find the paragraph beginning:** "The system implements a pre-approval validation mechanism"

**Edit boundary:** Replace Finding 2 only, including its continuation on p. 80. Keep the finding number; do not change respondent counts or survey findings here.

**Suggested replacement / insertion:**

The system includes exact identity checks, weighted Levenshtein-based record comparison and scholarship-commitment controls. In the inspected implementation, fuzzy matching supports duplicate-record review, whereas the current roster workflow checks identity and assignment conflicts and backend transactions restrict conflicting scholarship commitments. These mechanisms operate on the internal and externally sourced records actually stored in the system. Their demonstrated effectiveness must be reported using the documented test cases rather than inferred from their presence in the code.

**Exact location:** Chapter V - Conclusions > duplicate-validation paragraph, p. 81

**Find the paragraph beginning:** "The pre-approval validation mechanism supports the prevention of duplicate scholarship allocation"

**Precise paragraph locator:** Use the paragraph beginning 'The pre-approval validation mechanism supports the prevention of duplicate scholarship allocation' and ending 'one active scholarship per student.' Replace that paragraph only.

**Suggested replacement / insertion:**

The implemented identity-matching and scholarship-commitment mechanisms support the institutional one-active-scholarship policy within the records available to the system. Suspected duplicate identities, roster conflicts and existing commitments have distinct workflow effects; account activation is not equivalent to scholarship assignment. The implementation does not establish universal duplicate detection, and conclusions about its effectiveness must remain limited to documented test results and data coverage.

**Exact location:** Chapter V - Recommendations > student-record verification paragraph, pp. 82-83

**Find the paragraph beginning:** "The researchers recommend improving student"

**Edit boundary:** Keep the existing recommendation; it already acknowledges false matches and missing data. Insert the following sentence after its continuation ending 'inaccurate validation results.' on p. 83.

**Suggested replacement / insertion:**

Future validation should evaluate the configured similarity thresholds with labeled matching and nonmatching records and standardize scholarship closure behavior for inactive and terminated labels across workflows.

**Before submission:** Run the scenarios in R5 and attach actual evidence before turning implementation observations into claims of tested effectiveness. Keep the existing external-integration recommendation on p. 83. Update cross-references and page numbers only after the final manuscript is repaginated.

## What This Guide Does Not Change

The original thesis, adviser DOCX and application source remain unchanged. This guide does not resolve the adviser's separate recommendation-model, respondent-count, evaluation-calculation, security-evidence or research-gap comments. It identifies only the exact duplicate-detection and related consistency passages requested.